

FIELDCELL · DESIGN PRECIS

FieldCell design precis

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DOCUMENT CONTROL

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REVISION HISTORY

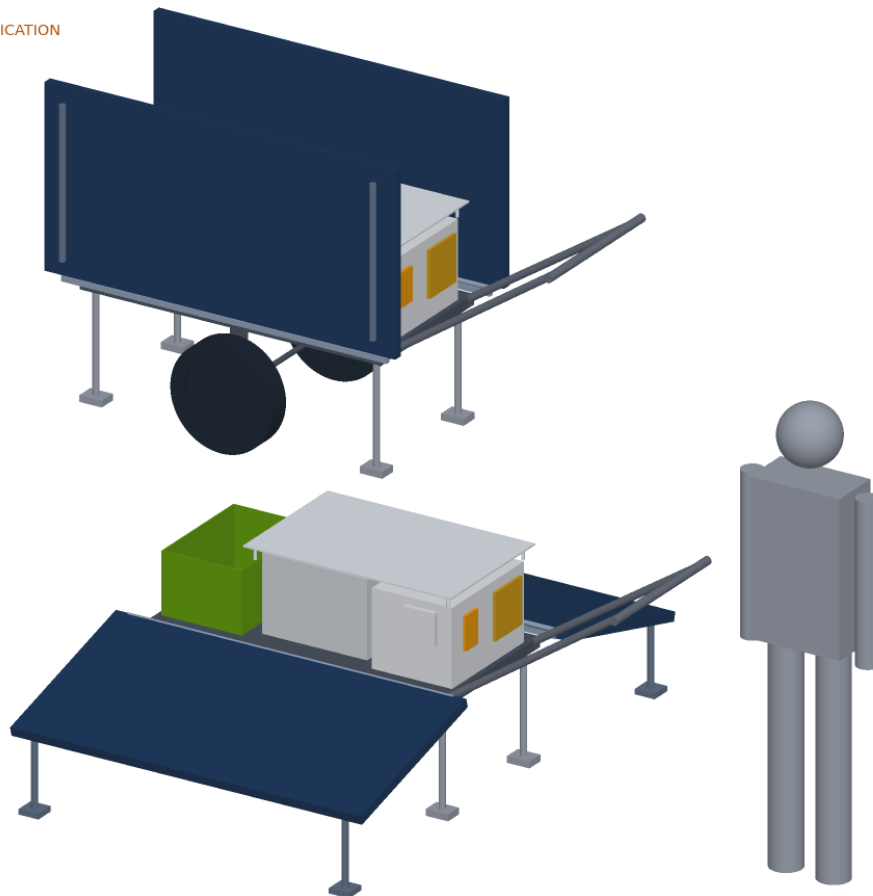
Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update. Decisions of FCL-DDR-001 recorded; numbers replaced by FCL-CAL-001; parametric model, drawing FCL-DWG-002 and priced BOM
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

FieldCell design precis

FieldCell is a two-wheel hand cart whose two sides are 200 W solar panels. For travel the panels stand up as the cart walls; on site they fold out as wings on outrigger legs. Between them sit a 1.28 kWh LiFePO4 pack, an MPPT charger, a 1 kW inverter and DC outlets. The TRL 3 calculations (FCL-CAL-001) show that it stores about 1.17 kWh a day from its own PV at 4 kWh/m²/day of global horizontal irradiation, carries a 0.9 kWh/day field load for 1.08 days without sun, and deploys in about 6 minutes. A reflective sun shade covers the battery and electronics enclosures, and the wing hinges sit on 20 mm spacers to clear the tyres. The parts cost about \$2,051 against a \$2,100 budget. One requirement is not met, narrowly: the cart weighs about 75.1 kg against a 75 kg limit (R6).

FieldCell

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. FieldCell deployed (front) and stowed for travel (behind), with a 1.75 m person for scale. Generated from the parametric model `cad/src/model.py`.

■ How it works

1. **Travel.** The wings are latched upright as the cart sides, over the battery and electronics. One person pulls or pushes the cart by the T-handle on two 16 in (406 mm) flat-free wheels. The sun shade sits on four short posts above the enclosures and stays in place in travel. The center of gravity sits about 55 mm on the handle side of the axle, so the handle carries a light, positive load.

2. **Park.** Four stand legs fold down so the deck sits level at 460 mm.
3. **Deploy.** Each wing unlatches and swings out on a continuous hinge, mounted on a 20 mm spacer along the side rail, coming to rest 15° below horizontal on two fold-down outrigger legs. With the cart's long axis north to south, one wing faces east and one faces west. In wind, the four outrigger feet are staked.
4. **Charge.** The panels are pre-wired in parallel to the MPPT charge controller, which charges the 25.6 V pack. No cable is connected on site.
5. **Power.** The operator switches on the battery isolator and the inverter. Loads plug into the 120 V AC outlet (GFCI protected) and the DC panel (USB-C PD and 12 V sockets) on the handle end of the electronics box. A shunt monitor shows state of charge. A thermostat runs an IP54 filter fan when the box is warm, and the shade keeps direct sun off both enclosures.

FieldCell: daily energy flow at 4.5 kWh/m²/day global horizontal (all values estimates, FCL-CAL-001)

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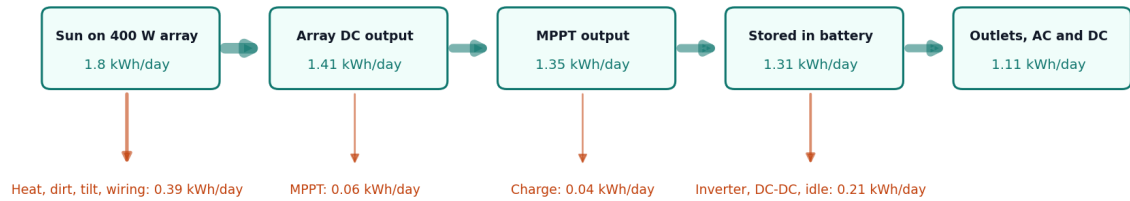


Figure 2. Daily energy flow at 4.5 kWh/m²/day global horizontal irradiation. All values are estimates from FCL-CAL-001.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`. Drawing FCL-DWG-002 gives the general arrangement and main dimensions.

Table 1. Main components.

#	Component	Choice	Notes
1	Cart frame and axle	40 x 40 x 1.5 mm steel square tube, 1,200 x 600 mm, expanded-metal deck, 20 mm axle	Welded steel, decided (FCL-DDR-001, D5); about 12.8 kg
2	Wheels (2)	16 in (406 mm) flat-free, 720 mm track	Flat-free, decided (D5); about 4 kg each
3	Handle	Steel T-handle, grip at 900 mm	Detaches for vehicle transport
4	Stand legs (4)	Folding tube legs with feet	Level the deck when parked
5	Battery enclosure	IP65 case, about 420 x 360 x 300 mm	Sits over the axle, lowest heavy item
6	LiFePO4 pack	25.6 V 50 Ah (1.28 kWh), 100 A BMS with low- and high-temperature charge cutoff	24 V system, decided (D1)
7	Electronics enclosure	IP54, about 320 x 460 x 300 mm, IP54 filter fan (about 40 m ³ /h) and exhaust filter	Outlet face toward the handle; thermostat filter fan decided (FCL-DDR-002)
8	MPPT charge controller	100 V, 20 A	At most 15 A from 400 W
9	Inverter	1 kW continuous, 2 kW surge, pure sine, high-frequency, 24 V input, 120 V 60 Hz	120 V first, 230 V 50 Hz variant documented, decided (D4)
10	DC outlet panel	2 x USB-C PD 100 W, 2 x 12 V sockets, 24 to 12 V 20 A converter	440 W DC in total
11	AC outlet with GFCI	GFCI duplex (120 V), in-use cover; 30 mA RCD socket for the 230 V variant	
12	Fusing and disconnect	Class T 100 A main fuse at the battery, isolator, DC breakers, shunt monitor	
13	PV wings (2)	200 W semi-flexible module on a ventilated aluminium frame, about 1,400 x 700 x 35 mm and 6.5 kg, 24 V class (Vmp about 36 V)	Parallel wiring, decided (D3)
14	Panel hinges, spacers and latches	Continuous hinge per side on a 20 x 20 x 2 mm aluminium angle spacer, over-center latches	Spacer raises the hinge line to 490 mm, decided (FCL-DDR-002); pinch points, see Safety
15	Outrigger legs (4)	Folding legs with stake loops and stakes, set the 15° tilt	Stakes required above about 9 m/s wind
16	Accessory and cable bin	Open bin, balances the electronics box	Cords, lights, stakes
17	Wiring and hardware	16 mm ² (6 AWG) battery cable, PV cable, glands, fasteners	Not modeled
18	Sun shade	Reflective aluminized fabric on a light aluminium frame, about 830 x 540 mm, four 50 mm posts, about 1 kg	Over both enclosures, decided (FCL-DDR-002)

■ Key numbers

All values are estimates from FCL-CAL-001 (docs/04-calcs/sizing.py), which checks every requirement.

Table 2. Key numbers.

Quantity	Value	Basis	Requirement
Stored energy	1.28 kWh nominal, 1.15 kWh usable	25.6 V x 50 Ah; 90 % depth of discharge	R1 met
PV energy into battery	1.17 kWh/day at 4 kWh/m ² /day, 1.31 at 4.5, 1.46 at 5 (25 °C air)	400 W x GHI x 0.729 (transposition 0.947, temperature 0.915, other losses)	R3 met
East-west penalty	About 14 % against one panel aimed south at latitude tilt	Clear-sky model at 15, 30 and 45° N	
Peak charge power	About 260 to 290 W	Noon, east-west wings	
Full recharge, 10 to 100 %, no load	0.79 day at 5 kWh/m ² /day	1,152 Wh / daily yield	R3 met
Reference load from battery	1,063 Wh/day	897 Wh at the outlets, Table 4	
Autonomy with no sun	1.08 days	1,152 / 1,063 Wh	R4 met
Battery current	46 A at 1 kW AC; 66 A with full DC; 98 A surge	24 V at the low end	R2 met
Total mass	About 75.1 kg	Table 5	R6 not met (0.06 kg over 75 kg)
Center of gravity	55 mm toward the handle from the axle, 564 mm above ground	Table 5	R6
Handle force, travel	33 N level; 14 to 52 N at a 5° pitch	CG offset x weight / 1,240 mm axle to grip	R6 met
Lateral static tip angle	32.6°	atan(360 mm half-track / 564 mm CG height)	R7 met
Pull force, 10 % grade	117 N (gravel) to 147 N (grass)	736 N weight x (grade + rolling resistance)	R7 at risk
Ground clearance	193 mm	Under the axle, legs folded	R7 met
Deploy and stow time	6.0 and 5.0 min	Table 6	R5 met
Deployed footprint	About 2.0 x 2.0 m including the handle	Model, wings at 15°	
Stowed size, handle off	1,400 x 784 x 1,190 mm	Model	
Wing to tyre clearance, deployed	28 mm	Model, 20 mm hinge spacer	
Electronics box at 45 °C ambient	About 53 °C with the fan and shade, 70 °C sealed and unshaded, at 1 kW	Section G of FCL-CAL-001	R9 at risk
Pack under the shade, noon, 45 °C ambient	About 48 °C (55 °C unshaded)	Section G of FCL-CAL-001	R9 at risk

Quantity	Value	Basis	Requirement
Wing lift-off wind speed, unstaked	About 9.2 m/s	Normal force coefficient 1.2 on 0.98 m ² , wing weight 64 N	R10 met only staked
Parts cost	About \$2,051 against \$2,100	bom/bom.csv	R12 met

PV yield

FCL-CAL-001 reads "peak sun hours" as daily global horizontal irradiation (GHI) and converts it to the wings with a clear-sky transposition model. At 15° the east-west wings collect about 0.95 of GHI, nearly as much as a horizontal panel, but about 14 % less than one panel aimed south at latitude tilt; the penalty is larger in winter at high latitude. With the cells at about 47 °C in 25 °C air, the 400 W array delivers about 0.78 of its rating times GHI at its terminals and 0.73 into the battery. At 4.5 kWh/m²/day that is 1.80 kWh at rating, 1.41 kWh at the array, 1.35 kWh after the MPPT and 1.31 kWh into the battery (Figure 2).

Load profile and days of autonomy

The reference load is a small field post: a responder team's comms and lighting, one laptop station, and occasional tool use.

Table 3. Reference load profile (assumed).

Load	Power	Hours per day	Energy at outlet	Path
LED area lighting	30 W	8	240 Wh	DC
Radio, phone and satellite messenger charging	30 W	8	240 Wh	DC
Laptop and Wi-Fi router	50 W	6	300 Wh	AC
Power tool or small pump	700 W	1/6 (10 min)	117 Wh	AC
Total			897 Wh	

Table 4. Energy drawn from the battery (assumed efficiencies, FCL-CAL-001).

Path	At outlet	Efficiency	From battery
DC (USB-C and 12 V converters)	480 Wh	0.93	516 Wh
AC (inverter), including 10 W idle for 7 h	417 Wh	0.92 plus idle	523 Wh
Standby and filter fan			24 Wh
Total	897 Wh	0.844 overall	1,063 Wh

With 1.15 kWh usable, the pack carries this load for 1.08 days with no sun. At 4 kWh/m²/day the PV stores 1.17 kWh/day, a surplus of about 10 %; at 4.5 the surplus is about 24 %. The system breaks even at about 3.6 kWh/m²/day. A run of cloudy days is not covered. Inverter idle draw is the most sensitive assumption: each extra 5 W costs about 3 % of the daily load.

Mass and center of gravity

Table 5. Mass budget, wings stowed, deck level. X is measured from the deck center toward the handle; the axle is at $X = 30$ mm.

Item	Mass (kg)	X (mm)	Z (mm)
Frame, deck and axle	12.8	0	430
Wheels, flat-free (2)	8.0	30	203
Handle	2.7	950	700
Stand legs	1.8	0	230
Battery enclosure	3.0	0	610
LiFePO4 pack	12.0	0	576
Electronics enclosure, filter fan, inverter, MPPT, fusing, monitor and outlets	11.3	420	568
Accessory bin and cables	4.0	-420	560
PV wings, stowed (2)	13.0	0	840
Hinges, latches, outriggers and stakes	3.0	0	780
Hinge spacers (2)	0.5	0	463
Sun shade	1.0	185	810
Wiring and hardware	2.0	200	600
Total	75.1	85	564

The center of gravity is 55 mm on the handle side of the axle. At the 900 mm grip the handle carries about 33 N (3.4 kgf), and it stays positive between 14 and 52 N if the user holds the cart 5° nose-up or nose-down. The load in the accessory bin is the trim: moving 4 kg of cords from the bin to the electronics end (840 mm) raises the handle force by about 27 N. Amish relaxed the R6 limit to 75 kg on 2026-09-25 to keep the decided steel frame and flat-free tyres (FCL-DDR-002). The hinge spacers and sun shade decided the same day add 1.5 kg, so the cart is 0.06 kg over the new limit, within the accuracy of the estimate; a shade of 0.9 kg or less would close it. A bolted aluminium frame would bring the cart to about 69.0 kg. How to close the gap is proposed in `docs/REVIEW.md`, awaiting Amish.

Deployment sequence

Table 6. Deploy and stow time estimates, one person.

Deploy step	Time	Stow step	Time
Turn the cart north to south, drop four stand legs	1.0 min	Loads off, inverter off, isolator off	0.5 min
Unlatch the first wing, fold it out, swing down its two outrigger legs	1.5 min	Pull four stakes	1.0 min
Same for the second wing	1.5 min	Fold the first wing's outriggers, raise and latch it	1.5 min
Stake the four outrigger feet (when wind is expected)	1.0 min	Same for the second wing	1.5 min
Isolator on, check the monitor, inverter on, plug in loads	1.0 min	Raise four stand legs	0.5 min
Total	6.0 min	Total	5.0 min

■ Key design choices

- **Panels as the cart walls, in fixed east-west wings at 15°.** The wings protect the battery and electronics in travel and need no separate storage, cables or aiming on site; they cost about 14 % of the energy a south-aimed panel would collect. Decided by Amish, 2026-09-25 (FCL-DDR-001, D2).
- **24 V class panels in parallel.** With Vmp about 36 V each, the panels can be paralleled into one MPPT and still charge a 25.6 V pack, so the off-sun wing does not limit the other. At 70 °C cell the headroom above the absorption voltage is under 1 V, so the chosen module must have Vmp of at least 36 V at STC. Decided by Amish, 2026-09-25 (D3).
- **25.6 V battery system.** Halves the current of a 12.8 V system (46 A instead of about 90 A at 1 kW), allowing thinner cable and a smaller fuse, while 24 V inverters and MPPT controllers remain common. Decided by Amish, 2026-09-25 (D1).
- **120 V 60 Hz output first.** A 230 V 50 Hz variant swaps the inverter and uses a 30 mA RCD socket. Decided by Amish, 2026-09-25 (D4).
- **Welded steel frame and flat-free tyres.** Easiest garage build and no punctures on debris, at a mass cost that now puts the cart over R6. Decided by Amish, 2026-09-25 (D5).
- **SwapCell compatibility as a future option only.** A later variant could take a SwapCell 48 V pack in place of item 6, which would need a 48 V inverter and charger. It would rely on three SwapCell interface v0.3 items approved portfolio-wide: a wake method for hosts without CAN, a charge-while-discharging mode (FieldCell charges from PV while loads run) and a latch vibration rating for vehicles. Decided by Amish, 2026-09-25 (D6).
- **Heavy parts low and over the axle.** The battery sits directly over the axle; the electronics box and the accessory bin balance each other fore and aft. Kept as a layout rule, decided by Amish, 2026-09-25 (FCL-DDR-002).
- **Forced ventilation of the electronics box.** A thermostat-controlled IP54 filter fan (40 to 60 m³/h) and exhaust filter keep the box within about 10 K of ambient at 1 kW; a sealed box would reach about 70 °C at 45 °C ambient. Decided by Amish, 2026-09-25 (FCL-DDR-002).
- **Sun shade over the enclosures.** A reflective shade on four posts keeps direct sun off the battery case and electronics box, holding the pack at about 48 °C instead of 55 °C at 45 °C ambient, for about \$30 and 1 kg. Decided by Amish, 2026-09-25 (FCL-DDR-002).

- **Hinge spacer.** A 20 mm aluminium angle under each hinge raises the hinge line from 470 to 490 mm and opens the deployed wing-to-tyre clearance from 8 to 28 mm. Decided by Amish, 2026-09-25 (FCL-DDR-002).
- **Irradiance basis.** "Peak sun hours" in R3 and R4 means daily global horizontal irradiation in kWh/m²/day. Confirmed by Amish, 2026-09-25 (FCL-DDR-002).

FieldCell: exploded view

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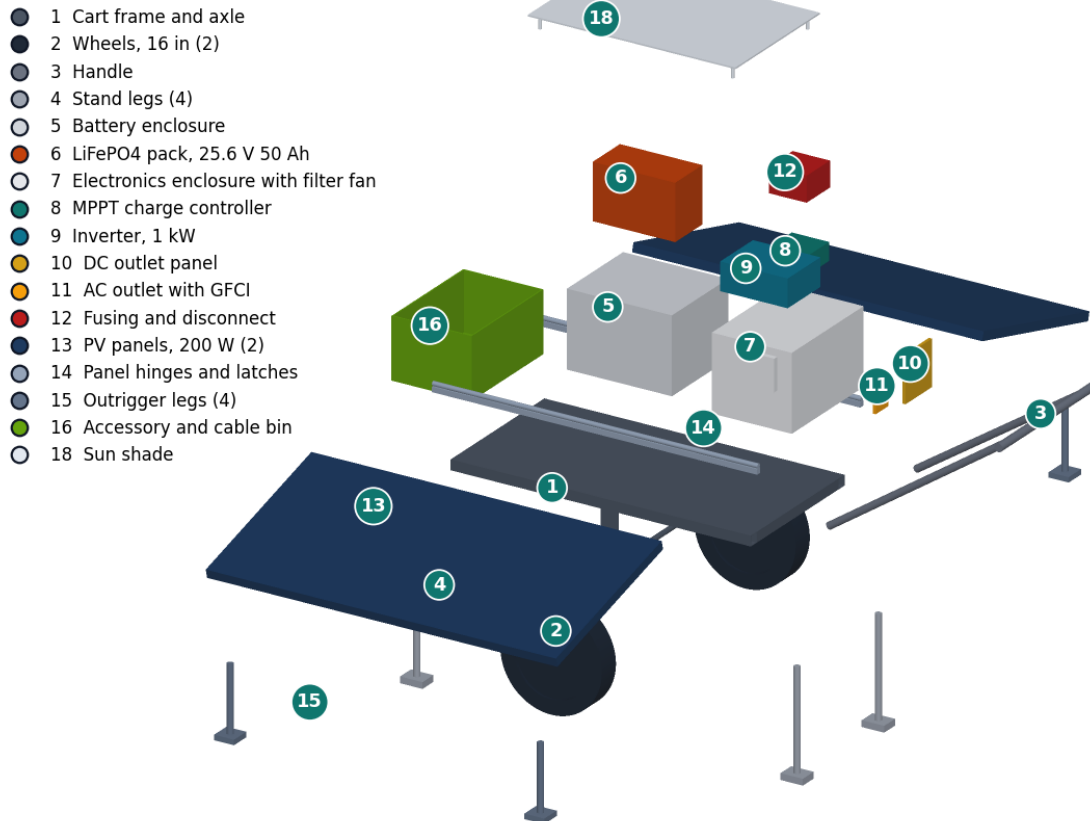


Figure 3. Exploded view with numbered callouts matching `bom/bom.csv` (item 17, wiring, is not modeled). Callouts 2, 4, 13 and 15 each label one of a pair; the unlabeled twin is the same part.

■ Safety

Safety: FieldCell combines a 1.28 kWh lithium battery, 120 V or 230 V AC output, high-current DC wiring, always-live PV panels and a cart of about 75 kg with folding wings. Every one of these is hazardous if the design or the build is wrong.

- **Lithium battery.** Use a LiFePO4 pack with a BMS that protects against overcharge, over-discharge, overcurrent, short circuit and over-temperature, and blocks charging below 0 °C and above about 45 °C. Put a class T fuse within about 150 mm of the positive terminal. Secure the pack so it cannot shift in a tip-over, keep the enclosure vented to outside air, and keep sharp objects and debris out of it. In noon sun at 45 °C ambient the pack can

reach about 55 °C unshaded and about 48 °C under the sun shade, so keep the shade in place. Shipping a pack of this size is regulated as dangerous goods.

- **AC output (120 V or 230 V).** Protect every AC outlet with a GFCI (120 V) or a 30 mA RCD (230 V). Bond the inverter neutral and ground to the frame only as the inverter maker specifies for a stand-alone portable source. Use in-use covers in rain, and keep outlets off wet ground.
- **Not for building wiring.** FieldCell has no power inlet and must never be connected to building or grid wiring, including by a "suicide cord" into an outlet. Back-feed can kill utility workers and damage both systems. Any such connection is out of scope.
- **DC wiring and fusing.** Fuse every circuit at its source: main class T fuse at the battery, breakers for the inverter, MPPT and DC outlets, and a PV breaker as the array disconnect. Size cables for the fuse rating. A battery isolator must be reachable without tools. The 2 kW surge with full DC load draws about 118 A for a few seconds; the BMS and fuse must tolerate this.
- **PV panels are live whenever lit.** Open-circuit voltage is about 45 V per panel, up to about 50 V in cold weather, and cannot be switched off at the panel. Do not disconnect PV connectors under load, and cover or disconnect the panels before working in the electronics box.
- **Lifting and handling.** At about 75 kg the cart must not be lifted by one person. Use two people to load it into a vehicle, and remove the battery pack (12 kg) first when possible.
- **Tipping.** The lateral tip angle is about 32.6° when stowed; do not traverse slopes across the fall line, and keep the handle low when descending. When deployed, wings can lift in wind above about 9 m/s unless the outrigger feet are staked; stow the wings in high wind. A stowed cart can be blown over at about 20 m/s side wind.
- **Pinch points.** The wing hinges, latches and folding legs can trap fingers. Latches must hold the wings firmly in travel. The deployed wing clears the tyre by about 28 mm; keep fingers out of that gap when folding.
- **Heat.** The inverter and MPPT shed up to about 140 W inside the electronics box. Keep the filter fan and exhaust filter clear, and do not run at high load with the vents covered or with the sun shade removed.

■ Open questions

- Mass: 75.1 kg against the relaxed 75 kg limit (R6), 0.06 kg over. How to close the gap is proposed, awaiting Amish (docs/REVIEW.md).
- Inverter derating data at 45 °C (R9) and supplier quotes for the pack, PV wings and inverter to confirm the \$2,100 budget. Decided by Amish, 2026-09-25 (FCL-DDR-002); to be obtained with supplier selection.
- Confirm the inverter idle draw and fan noise (R11) and the filter fan noise from datasheets.
- Whether an AC charger or a vehicle 12 V input should be added for cloudy periods. Open option, awaiting Amish (FCL-DDR-001, O3).
- How FieldCell relates to PowerBox and SwapCell in the portfolio, so the three do not overlap. Awaiting Amish (O2).
- Handle geometry and grip height for users from 1.55 to 1.90 m tall.

Concept media: [blueprint sheet](#), [cutaway](#), [interactive 3D model](#). General arrangement: FCL-DWG-002.